Analysis of "No Signal to Rule Them All" and Associated Network Concepts

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1. Core Paper Analysis (What We Did)

To begin with, we looked at the manner in which the research community quantifies whether CCAs which utilized INT signals out perform E2E signals.

- **The AI Generator (R+):** The application of Remy had been changed by the Research group, and this installation of Remy had been modified by the researchers into R+ to allow the researcher to create a method for producing custom(s) CCAs using a Match-Action table that relates Signal Space to Actions.
- **The Trade-off Paradigm:**
  - **Link Signals:** While using link signals the objective is to keep the total utilization of the link as close to 100% as possible, thus maximizing the maximum amount of queue delay to provide no queue delays that maximize queue delays (or produce a lower max throughput).
  - **Queue Signals:** The objective is to keep the size of the queue as small as possible, thus producing maximum throughput with some queuing delay.
- **Simulation Environments (ns-3):** we could decompose the test network topologies into a "Dumbbell" (one bottleneck) and a "Line" (many bottlenecks) and confirmed that the links between the router and host were over utilized, leading to congestion being concentrated in the links between routers.
- **Data Analysis (Figures 6-9):** By reviewing the CDF Plots and correlation plots, we determined that there is a statistically significant difference in utility weightings of absolute scores with respect to E2E at INTs of 100d or 2t; however, the improvement between pure scores at E2E and INT is less than 10%.
- **The Secondary Metric Advantage:** In our evaluation, we found no improvement over E2E signaling concerning primary priority metric. However, the strength of INT comes from the effect it has upon secondary metrics only.
- **Reverse Path Congestion:**We examined the effects of excessive traffic flowing backward and how that created E2E delays (as a build-up) and how with INT there would be a distinguished advantage because they do not consider cumulative metrics but utilize mini-max metrics per hop only.

- **The BBR Proof:** In our study of human-created BBR (BBR') we saw that the algorithms used (or not used) in modifying the human BBR algorithm also had similarly impressive results with E2E signaling, although less so when used as E2E signaling (RTT). The overall findings indicate that the types of signals do not have nearly the value that the algorithms do.

2. Supplementary Network Architecture (What We Read Extra)

To fully comprehend the results contained within this document, we also performed an analysis of some foundational networking techniques that had either been evaluated by the authors as alternatives to their study or utilized in the study itself.

- **RTT Mechanics:** A detailed analysis of how RTT (Round Trip Time) is first calculated during the TCP 3-way handshake (SYN/SYN-ACK), examining propagation delay, serialization delay, queuing delay and processing delay.
- **Explicit Congestion Notification (ECN):** In-depth analysis of how routers set the 11 bits of the IP header to indicate a buffer is approaching capacity (marking) prior to overflowing their buffers and how this marking is returned to the sender (by the receiver) using the ECE flag in the TCP header.
- **Datacenter TCP (DCTCP):**Analysis of the issue of microbursts in the datacenter where DCTCP proposes to address this issue via the transition from the standard TCP's binary halving mechanism to a proportional increment (or decrement) of the window size based on the percentage (α) of ECN-marked packets received (by the sender).
- **Utility Function Mathematics (U):** We clarified the scoring formula:

  $$U = c_t \cdot \log_2(\text{throughput}) - c_d \cdot \log_2(\overline{\text{delay}})$$

  Analysis of the mathematical formulation of the score, where the function is made more important by the use of a higher value for cd, thus forcing the application to maintain strictly empty queues.
- **INT Hardware implementation:** We showed the way the system updates its telemetry and how routers do MAX aggregation, and all of the routers will write over the packet header of a packet that is 8 bytes long with the new value of the packet that they received from the router that has the highest local queue size.

3. Practical Implementation and Execution (How We Ran the Code)

To prove out the theories we built and simulated R+, the Remy simulator, a learning simulator that uses artificial intelligence. Since there is a great deal of computing power required to train an AI, we configured the simulations for running localized simulations.

- **Environment Optimization (The "Fast Run"):** The original breeder generates several millions of configurations across several different networks. Using the `src/configrange.cc` source file, we modified how this is performed so that the training can all take place within the same network. We accomplish this by making the `ConfigRange` class only test one fixed environment (by forcing all links to use the same upper bound and lower bound values).

- **Resolving Compilation Constraints:**In addition to reducing the number of simulation ticks from `1,000,000 to 10,000`, this change caused an error to be raised because of the C++ compiler's strict implementation of the `-Werror=reorder` rule. An important observation regarding this rule was that all of the initialization of class member variables defined in a `.cc` file must be done in exactly the same order as those same member variables are declared in a .hh file.

- **Compilation for Performance:** Because of the iterative nature of genetic algorithms, normal debug modes do not work well with genetic algorithms; thus, we must use the `-O3` compiler flag (`make clean && make CXXFLAGS="-O3"`) to allow this to be successful.

- **Executing the Training Loop:** We successfully ran a localized training loop using the compiled `remy` binary. We restricted the execution to a single generation to quickly verify the rule-generation logic:
`./src/remy nsrc=2 cf=configs/config-default.cfg of=fast_rules.dna gen=1`
This command successfully evaluated the baseline rules and generated an initial `WhiskerTree` ruleset tailored to our controlled network limits.

4. Final Conclusions Drawn

1. **"No Signal Rules Them All":** In most cases, well-resolved E2E measurements (e.g., round-trip times to the nearest millisecond and receptions up to 100%) in a "typical" network will provide a mathematically sufficient amount of measurement data. All will meet performance requirements of being within ±3% of performance from well-resolved INT measurements created from switches.

2. **Logic > Telemetry:** A Congestion Control Algorithm's success depends heavily on how its control loop is designed, specifically which conditions determine when to trigger or probe future states (of congestion), more so than how accurately the algorithm receives the network signals used to input into it.

3. **The Niche for INT:** In-Network Telemetry is only strictly necessary in extreme edge cases:

    - When the utility priority is heavily skewed toward one metric (e.g., zero-tolerance for delay).
    - In highly congested, multi-hop environments where reverse-path traffic corrupts standard RTT measurements.

4. **Hardware ROI:** For network engineers, upgrading an entire fleet of switches to support hop-by-hop INT stamping is likely a poor return on investment for general

traffic, as standard E2E metrics can be tuned to deliver nearly identical throughput and delay profiles.